BRAC UNIVERSITY

Merul Badda, Dhaka, Bangladesh

CSE221 : Algorithms

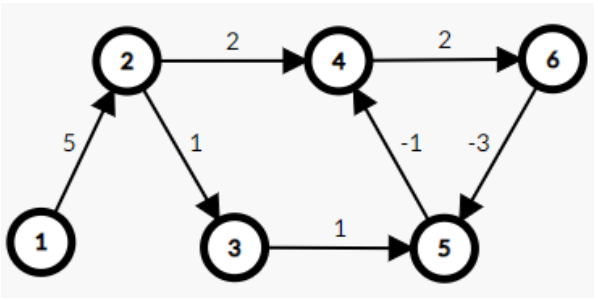
Quiz 3

Summer 2024

Duration: 20 minutes **Total: 15 marks**

Name:	**ID:**	**Section:**

1. Simulate the suitable SSSP algorithm on the following graph. Write the time complexity of your proposed algorithm as well. (8 marks)

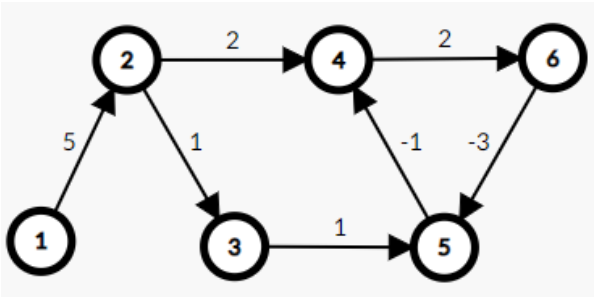


Solution:

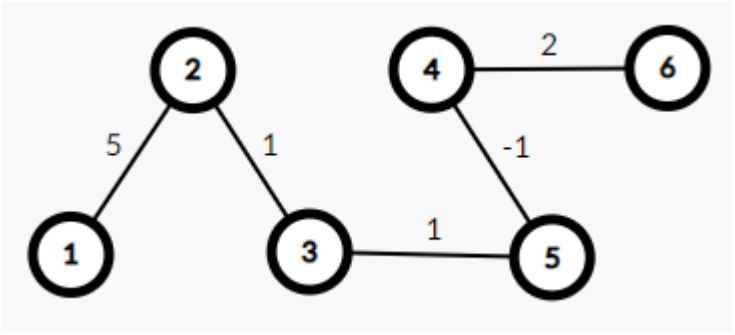
Node	0	1	2	3	4	5
1	0	0	0	0	0	0
2	inf	5	5	5	5	5
3	inf	inf	6	6	6	6
4	inf	inf	7	7	6	5
5	inf	inf	inf	7	6	6
6	inf	inf	inf	9	9	8

TC → O(VE)

2. Simulate the Prim's algorithm for finding MST on the following graph. Write the time complexity of the algorithm as well. (8 marks)



Solution:



TC → O(VlogV + ElogV)